

EXPERIMENT 25

Create and Manage a GitHub Repository Using Git

1. Aim

To create a GitHub repository with a README.md file, clone it to a local machine, edit the README, and push the changes back to GitHub.

2. Create GitHub Repository

Create a new repository on GitHub and select Initialize this repository with a README.

3. Clone the Repository

```
git clone https://github.com/<username>/<repository>.git
cd <repository>
```

4. Edit README.md

```
# My Personal Project
```

This project is created as part of my software engineering work.
It contains the project source code and documentation.

Save the changes to README.md.

5. Stage, Commit and Push

```
git status
git add README.md
git commit -m "Update README project description"
git push origin main
```

6. Verification

After pushing, refresh the GitHub repository page and verify that the updated README and the new commit are visible.

7. Testing Screenshot

The screenshot below shows the GitHub repository's commit history on the main branch, providing evidence of changes being committed to the repository.

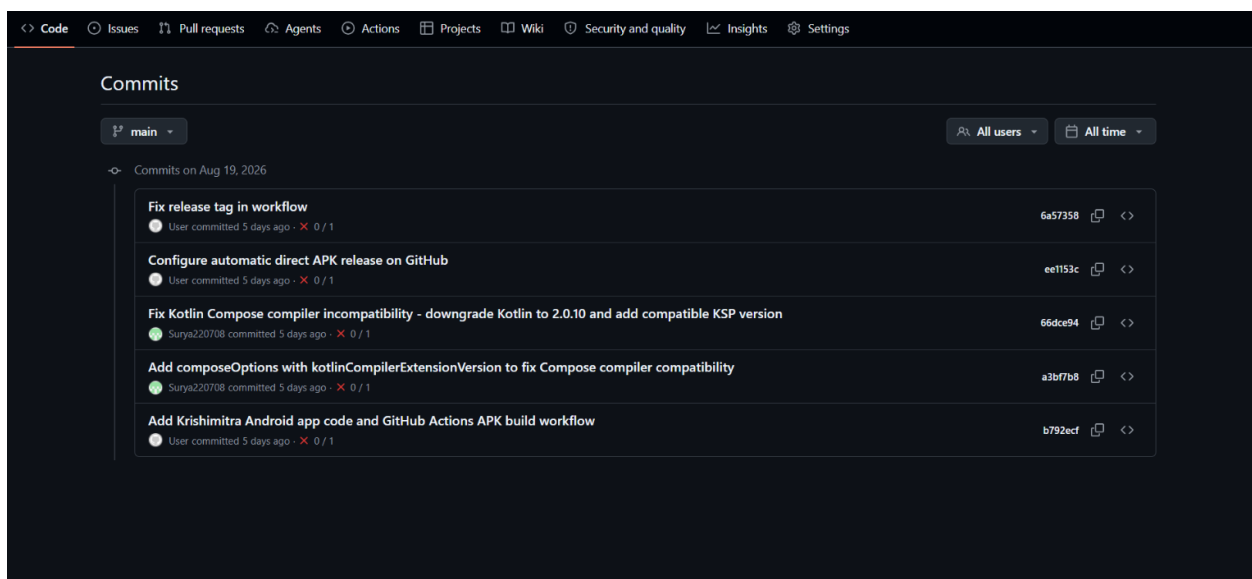


Figure 1: GitHub repository commit history showing pushed changes.

8. Result

The GitHub repository was created and cloned successfully. The README.md file was edited, the changes were staged and committed, and the updated commit was pushed to GitHub.

9. Conclusion

Git and GitHub provide an effective workflow for maintaining project files, tracking changes, and synchronizing local work with a remote repository.